ARMSTRONG

ZL-9001A

Computer Controlled Carton Compression Testing Machine



Product illustration. Confirm the supplied configuration and accessories in the quotation.

MODEL	CAPACITY / CONFIGURATION
ZL-9001A	1,000 kgf load cell / 800 x 800 x 800 mm

Computer-controlled compression frame with a servo drive, low-backlash twin ball screws and guide columns. TM2101 software handles test setup, measurement curves and reporting.

Supplied and supported by Hung Ta Instrument.

Applications and specifications

Determine the load-bearing behaviour of corrugated cartons and packaging containers in compression or holding tests. The source references GB/T 4857.4-92; confirm the test method and edition required by the customer.

Parameter	Specification
Load cell rating	1,000 kgf (approximately 9.81 kN); confirm maximum operating load
Compression space	800 x 800 x 800 mm
Drive	Servo motor with twin ball screws and guide columns
Speed accuracy	Within +/-0.2% as stated in the source; confirm verification conditions
Control software	TM2101 on a compatible computer
Supply	Single phase, 220 V, 50 Hz; confirm the nameplate

For quotation, provide the specimen material and size, test method, required force range, fixtures and report requirements. Warranty and service are governed by the Hung Ta quotation.

Technical reference

Operating and software guidance. Illustrations may show example readings or a different software revision. Use the instructions for the delivered unit.

2 Safety precautions

Safety signal words

Observe the following Danger, Warning, Caution and Note instructions to prevent accidents and hazards when using this instrument.

DANGER | Failure to follow the instruction may cause injury to the operator.

WARNING | Failure to follow the instruction may damage the instrument.

CAUTION | The instruction concerns the quality and accuracy of test results.

NOTE | Additional guidance for operating the instrument.

Safety symbols on the instrument

Safety symbol	Meaning
Warning	Consult the operating manual where this symbol appears.
Hazardous voltage	High-voltage hazard.
Protective earth	Identifies the protective earthing terminal.

3 Machine appearance

Reference photograph. The actual supplied machine configuration takes precedence.

4 Machine operating instructions

4.1 Specimen preparation

Prepare specimens according to the applicable product test standard and your test requirements.

4.2 Checks before testing

Before each test, confirm the following:

1. Confirm that the power supply meets the specified requirements.

2. Set the upper and lower safety limits to appropriate positions.

4.3 Test procedure

1. Connect the power supply specified in the source procedure: AC 220 V, 3 A. Verify the actual machine nameplate before connection.

2. Turn on the power switch.

3. Adjust the upper and lower limit collars to suitable positions and tighten them with a hex key.

4. Enter the required test parameters on the test screen and position the specimen securely.

5. Run the compression test.

6. Repeat for the next specimen or switch off the power.

7. Record the test results when testing is complete.

EMERGENCY STOP | In an emergency, use the machine stop or emergency stop control to stop testing. Identify the installed emergency stop before operation.

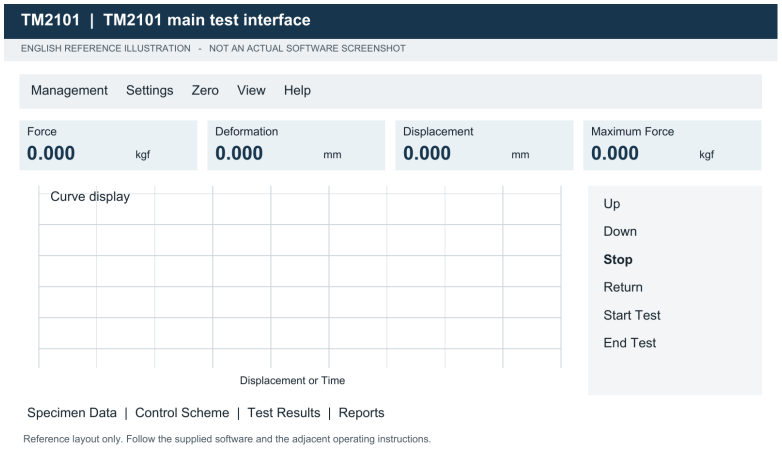
If abnormal operation occurs, stop the machine immediately and isolate the power where necessary. The operator must remain at the machine during testing.

TM2101 testing and reporting

The English interface brings specimen data, control schemes, test curves and results together. Hung Ta can assist with software installation, commissioning, verification and report setup for the supplied system.

Main interface

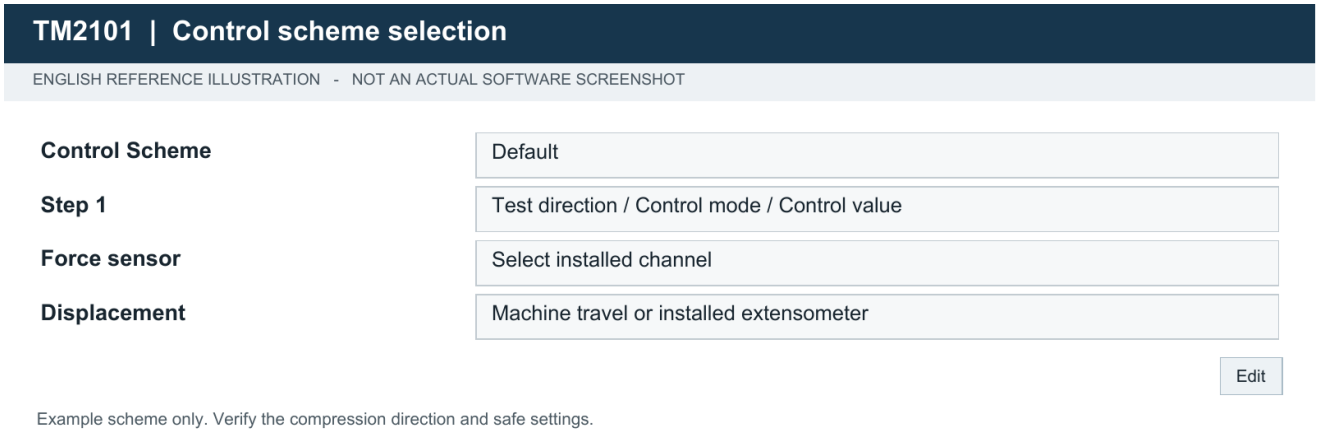
View force, time, deformation, displacement and the selected test curve. Use the motion and test controls for the configured machine.



English interface illustration. Display depends on the installed software version.

Specimen and test setup

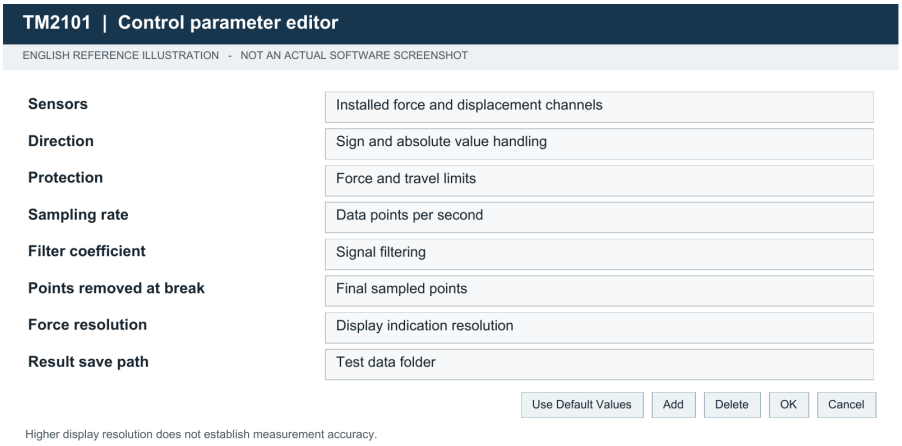
Select a stored specimen or enter a new one. Choose a compression or constant-force holding method suited to the applicable packaging test.



English interface illustration. Display depends on the installed software version.

Control parameters

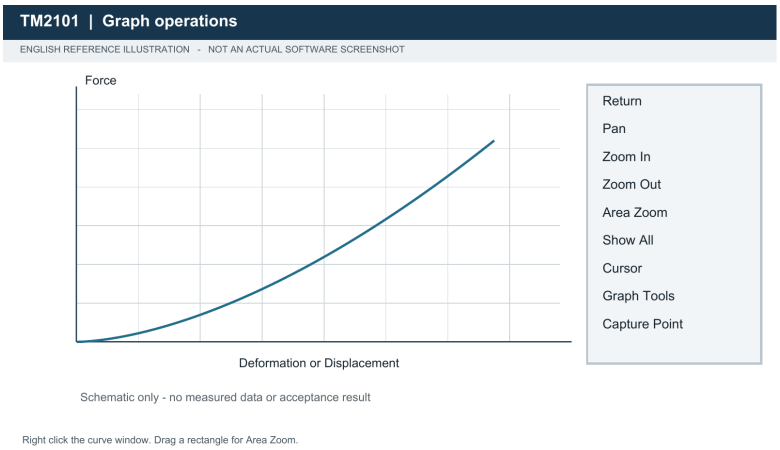
Set applicable directions, speed, limits, holding conditions and sampling settings within the approved machine configuration. Verify the required test method with Hung Ta before setup.



English interface illustration. Display depends on the installed software version.

Curve review and points

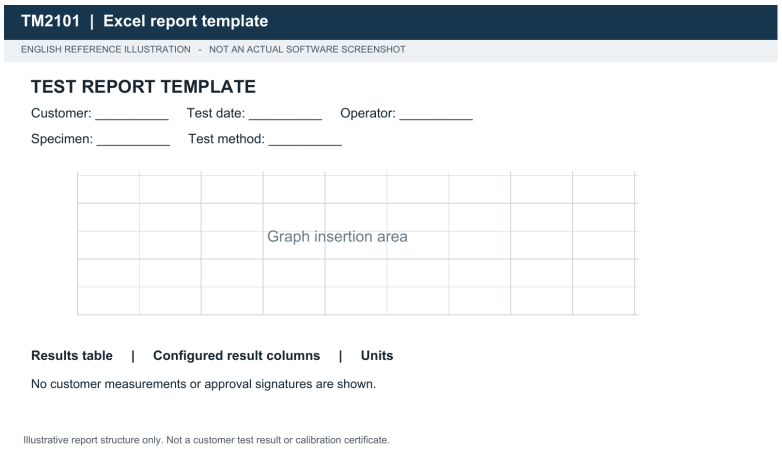
Review the force/displacement curve, compare selected results and capture relevant points. The software supports configurable results and curve annotations.



English interface illustration. Display depends on the installed software version.

Word and Excel reports

Select report fields, units, test results and curves. Use the configured Word or Excel template to record specimen and test information.



English interface illustration. Display depends on the installed software version.

8 Maintenance

1. Install the instrument on a firm, stable supporting surface before use.
2. Do not move or handle the machine during operation.
3. Use the specified supply voltage. Excessive voltage can damage components.
4. Contact Hung Ta service support promptly if the instrument operates abnormally.
5. Provide adequate ventilation while the machine operates.
6. Lubricate mechanical components at appropriate intervals because they operate under substantial loads.
7. Clean the machine after each test and keep it free of debris.
8. Wipe the control box with a dry cloth, not a wet cloth.
9. Ensure the equipment is properly earthed.

Service and support

For technical support, provide the ARMSTRONG model, serial number and a description of the issue to Hung Ta Instrument.